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CONTEXTUALITY AS AUTOMATA: OPEN GENERATORS AND THE DYNAMICS OF QUANTUM PHENOMENA

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Context and Motivation

CHAPTER 1

1.1. *Why start from observation?* ---

Scientific theories do not begin with reality given all at once. They begin with phenomena: outcomes, traces, correlations, and reports produced through concrete experimental interfaces. A hidden variable, a state, a Hilbert space vector, or an automaton is therefore not direct access to reality itself. It is a model introduced to organize what is observed.

This distinction is modest, but it is important for the rest of the report. We do not try to reveal a final hidden world. We ask which hidden structures can explain the visible data, how much structure they require, and what they still fail to explain. In that sense, the word **model** should be read operationally. A model is a disciplined way of connecting observed phenomena to a possible hidden organization.

This position is not meant to weaken science. On the contrary, it makes the task more precise. Since the hidden structure is not directly given, it must be reconstructed under constraints. It must fit the observations, respect the operational limitations of the experiment, and use a controlled amount of structure. A model is then judged by its explanatory power, by its coherence, and by the resources it needs.

1.2. *Truth, understanding, and interruption* ---

In this report, truth is used in an operational and provisional sense. A model is accepted when it is the best available explanation of the observed phenomena under explicit constraints. If the observations change, or if the class of admissible models changes, the accepted model may change as well. This does not make truth arbitrary. The choice of a model is constrained by data, by mathematical consistency, and by the cost of the hidden resources used in the explanation.

Understanding is slightly different. A model may give correct predictions while remaining too opaque to reuse. Here, a model is understood when it can be compressed, transmitted, interrupted, and reconstructed without losing its explanatory role.

A simple analogy is curve fitting. Suppose that an experiment produces a sequence of points (x_i, y_i) . In a classical analysis, one may fix a class of models, for instance polynomials of degree at most 3, and choose the polynomial that best fits the first n points: One then says that the model is good if the final error is small enough.

This report asks for a more dynamic requirement. The explanation procedure itself should be observable and interruptible. After each new point, we obtain a new model, a new error, and a transition from the model at step n to the model at step $n + 1$. A

good explanatory process should not only end with a small error. Its successive updates should also follow a coherent pattern.

This is the sense in which we want a kind of meta-model. We do not only want a final explanation of the completed experiment. We also want to track the history of partial explanations, their errors, and their possible continuations. If the experiment is interrupted at time n , the current state should still have a meaning.

This point will become central later. A probability table is a completed summary. It has already forgotten the order of events, the number of occurrences before normalization, and the possible unfinished mechanisms that produced the data. If our aim is only prediction, this loss may be acceptable. If our aim is reconstruction, it is too strong. We need a language that keeps enough information to restart the process after an interruption.

The report therefore takes seriously a simple methodological principle: before normalizing, count; before completing, keep the prefix; before replacing a mechanism by a probability distribution, ask what hidden process could have produced it.

1.3. *Existing routes toward the quantum model* —————

The question of which model best explains quantum phenomena is not new. Bell's theorem showed that local hidden-variable models cannot reproduce all quantum predictions [Bel64]. The CHSH inequality gave a particularly clear experimental and mathematical form of this obstruction [Cla+69]. These results changed the status of hidden mechanisms: the difficulty is not merely that the hidden variables are unknown, but that some families of local observations cannot be glued into a single classical global assignment.

One response is to search for general operational principles that single out quantum theory. Generalized probabilistic theories provide a broad framework in which classical theory, quantum theory, and many non-classical theories can be compared [Bar07]. In this setting, states, effects, and transformations are not assumed to be Hilbert-space objects from the start. They are abstract operational objects constrained by axioms. Several reconstruction programs follow this direction. Hardy derived quantum theory from a small set of reasonable axioms [Har01]. Chiribella, D'Ariano, and Perinotti derived the finite-dimensional formalism from informational principles, with purification playing a central role [CDP11]. Masanes and Müller gave another derivation from physical requirements concerning preparations, transformations, and measurements [MM11].

These works are important because they show that the Hilbert-space formalism is not the only possible starting point. One may instead ask which operational or informational constraints force that formalism. They also show how hard the problem is. If the axioms are too weak, one obtains theories larger than quantum theory. If they are too strong, the reconstruction risks hiding the quantum structure inside the assumptions.

The example of non-signaling correlations makes this tension concrete. Popescu and Rohrlich showed that there are non-signaling correlations stronger than quantum

correlations [PR94]. Thus no-signaling alone does not characterize quantum theory. The quantum set lies between the classical local polytope and the larger non-signaling polytope. The problem is then to explain why nature seems to choose this intermediate region.

There is also a second difficulty. Even inside the mathematical formalism of quantum correlations, different completions of the model are not obviously equivalent. One may use finite-dimensional tensor-product models, infinite-dimensional limits, or commuting-operator models. The relation between these descriptions is connected to Tsirelson’s problem and to Connes’ embedding problem [Jun+11].

The distinction has a physical meaning. In a finite tensor-product model, Alice and Bob are described by two separate Hilbert spaces, and the joint system is represented by $\mathcal{H}_A \otimes \mathcal{H}_B$. This is close to the usual laboratory picture of two separated devices. In the commuting-operator model, one uses a single Hilbert space $\mathcal{H}$ and only requires Alice’s and Bob’s observables to commute, for example $A_x B_y = B_y A_x$. This algebraic condition says that the two measurements are compatible: changing their order does not change the predicted statistics. It is natural in algebraic quantum field theory, where observables localized in spacelike separated regions are required to commute. However, it is less directly tied to a finite experimental mechanism than a concrete tensor decomposition.

The issue can be stated in terms of correlation sets. Let C_q denote the set of correlations obtained from finite-dimensional tensor-product strategies. Let C_{qs} denote the spatial tensor-product model, where the Hilbert spaces may be infinite-dimensional. Let $C_{qa} := \overline{C_q}$ denote the approximable quantum set, obtained by closing the finite-dimensional set. Finally, let C_{qc} denote the commuting-operator set. These sets satisfy the inclusions

$$C_q \subseteq C_{qs} \subseteq C_{qa} \subseteq C_{qc}.$$

The inclusions are not only technical details. Slofstra showed that finite-dimensional quantum correlations need not form a closed set, so one can have $C_q \neq C_{qa}$ [Slo19]. Tsirelson’s problem asks, in one standard formulation, whether the approximable model and the commuting-operator model coincide, that is, whether $C_{qa} = C_{qc}$. The theorem $\text{MIP}^* = \text{RE}$ implies a negative answer: there are non-local games whose commuting-operator value cannot be approximated by finite-dimensional tensor-product strategies, hence $C_{qa} \subsetneq C_{qc}$ [Ji+22].

The zero-gap version of this story is also relevant. In the usual gap formulation of MIP^* , a verifier must distinguish a yes-instance from a no-instance with a fixed separation between acceptance probabilities. In MIP_0^* , this uniform gap is removed. The question becomes exact: is the entangled value equal to 1, or only strictly smaller than 1? This looks like a small change, and it is operationally natural when one cares about perfect realizability rather than an arbitrary error threshold. However, it changes the behaviour of the associated classes in a strong way. Mousavi, Nezhadi, and Yuen show this for zero-gap MIP^* , and their Figure 1 gives a useful picture of the different inclusions between the classical, entangled, and commuting-operator variants [MNY20]. Related work on commuting-operator formulations leads to a different computability-theoretic

behaviour again [Lin25]. Thus even a small modification of the rule, such as removing the promise gap, can change the model at a structural level.

These results do not decide which mathematical completion is physically real. They show a more modest but important point: the word “quantum” does not determine a unique operational object unless the allowed idealization is specified. A finite experiment gives finite data, with finite precision. It can rule out restricted finite-dimensional models, or give lower bounds on dimension, but it cannot by itself certify an exact infinite-dimensional Hilbert space or a genuinely non-tensor commuting-operator realization. Thus the separation $C_{\text{qa}} \subsetneq C_{\text{qc}}$ is mathematically decisive, while its direct physical interpretation remains more delicate.

This motivates the constructive stance of the present report. The aim is not to decide the ontology of Hilbert spaces. It is to work at the level where the experiment is actually recorded: visible events, integer counts, compatibility equations, and finite generators. A property proved at this level has a direct operational meaning before any passage to closures, limits, or operator-algebraic completions. In this sense, staying close to the experimental record is not only a simplification. It is a way to separate physically accessible structure from structure introduced by the mathematical completion of the model.

1.4. *From state spaces to constructive countings* —————

Most operational reconstructions start with a space of states, a set of effects, and rules for composing systems. This is natural when one wants to reconstruct the full probabilistic formalism. In this report, we take a different starting point. We begin with finite observations and integer counts.

The reason is simple. A normalized probability distribution erases the level at which it was obtained. For example, consider two possible events A and B . The distribution $\mathbb{P}(A) = 1$ and $\mathbb{P}(B) = 0$ may come from the count $(1, 0)$, from the count $(1000, 0)$, or from any count $(t, 0)$ with $t > 0$. After normalization, all these situations become identical. Empirically, however, they are not the same: one observation and one thousand observations do not carry the same amount of evidence. Keeping integer counts preserves this scale. It also makes it possible to ask which finite pieces of data are already compatible, which are not yet complete, and which hidden generator could still close them.

This is where contextuality enters. In the sheaf-theoretic approach, contextuality is an obstruction to gluing local data into a global section [AB11]. This language is well suited to our purpose because it separates local observation from global explanation. In the discrete version used here, local probability tables are replaced by integer count tables. Compatibility becomes equality of marginal counts on overlaps. Noncontextuality becomes the possibility of reconstructing the compatible count as an integer sum of deterministic global assignments.

The change from probabilities to counts is not only cosmetic. It turns convex geometry into semigroup geometry. Instead of asking only whether a point lies in a convex hull,

we ask how an integer count decomposes into irreducible generators. The Hilbert basis of the relevant semigroup then gives candidate elementary bricks of explanation. In CHSH, these bricks include local deterministic generators and lifted PR-type generators. This does not mean that PR boxes are accepted as physical final states. It means that they appear as elementary algebraic pieces of the compatible counting structure, and therefore must be interpreted.

1.5. Quantum phenomena as unfinished stabilization —

This perspective gives a useful way to read quantum phenomena. A quantum state will not be treated here as a primitive object whose meaning is already fixed. We instead ask whether some quantum features can be understood as effects of projection and incomplete stabilization. Superposition suggests that several hidden histories may give the same visible phenomenon. Entanglement suggests that local visible pieces may fail to factorize because they still come from one common hidden process.

This is deliberately only a guiding interpretation. The mathematical part of the report will be more modest. We start from finite observations, integer counts, and compatibility constraints. We then study the generators of these counts and the automata that describe their interrupted production. A completed compatible count is a stabilized object. A prefix is not an error; it is a partial observation that may still have possible completions. The residual future of an open generator records exactly what is missing for closure.

The aim is not to replace Hilbert spaces. It is to build a small discrete language in which such ideas can be tested. Contextuality is used as a laboratory for a broader question: how can visible phenomena be reconstructed from hidden processes without pretending that the hidden process is directly given?

Technical Preliminaries

CHAPTER 2

In this section we will introduce the technical concept from generic contextuality to our theory of discrete contextuality.

2.1. What is contextuality ? ---

Contextuality is the failure of a family of local observations to come from one global assignment. In a measurement scenario $\langle X, \mathcal{M}, O \rangle$, each context $C \in \mathcal{M}$ specifies a set of measurements that can be performed together, and a local outcome is a section $s : C \rightarrow O$. A classical deterministic explanation would require a single global section $g : X \rightarrow O$ whose restriction $g|_C$ explains the outcome seen in every context. Thus contextuality is not merely randomness: it is an obstruction to gluing compatible local descriptions into one global description.

This report uses this standard idea in a discrete form. Instead of starting from normalized real probabilities, we keep integer counts. A local table is therefore not only a point of a convex polytope, but a finite record of occurrences. Compatibility becomes equality of marginal counts on overlaps, and contextuality asks whether these compatible integer data can be reconstructed from deterministic global sections.

2.2. Introduction ---

In the following, we will use the same notation that in [\[AB11\]](#). We will add our theory on the top of this formalism and we will discuss how to consider discrete contextuality with this notation.

The goal is to model the experiment itself, rather than only the final probability distribution. We consider one or several laboratories equipped with measurement devices. Let X denote the set of available measurements, let $\mathcal{M}$ denote the set of measurement contexts, i.e. the subsets of measurements that can be performed jointly, and let O be a finite set of possible outcomes. An experimental run consists in choosing a context $C \in \mathcal{M}$ and observing an outcome assignment $s : C \rightarrow O$. Thus the primitive observational events are pairs (C, s) , where C is the performed context and s is the observed local outcome.

For example, in the classical $(2, 2, 2)$ Bell scenario, there are two parties, Alice and Bob, each with two possible measurements. We write

$$X = \{a_1, a_2, b_1, b_2\},$$

where a_1, a_2 are Alice's measurements and b_1, b_2 are Bob's measurements. The jointly performable contexts are

$$\mathcal{M} = \{\{a_1, b_1\}, \{a_1, b_2\}, \{a_2, b_1\}, \{a_2, b_2\}\},$$

and the outcome set is

$$O = \{0, 1\}.$$

This expresses the operational constraint that Alice and Bob can each choose one measurement per run, but Alice cannot perform both of her measurements simultaneously, so $\{a_1, a_2\} \notin \mathcal{M}$. Likewise, $\{b_1, b_2\} \notin \mathcal{M}$.

In this sense, the scenario specifies not only which outcomes may be observed, but also which observations can coexist in a single experimental run.

The technical point of this section is the following. A context table is only local data. A genuine empirical model is obtained only when all context tables can be compared on overlaps and when they have a common level. Therefore we must distinguish three objects: a local counting N_C , a family $N = (N_C)_{C \in \mathcal{M}}$, and the corresponding vector $N \in \mathbb{N}^V$ indexed by visible contextual events. The notation is intentionally the same, but the type of the object changes with the context.

As an illustration, consider the following probability table for a quantum strategy in the CHSH scenario. The rows correspond to the measurement settings (x, y) chosen by Alice and Bob, and the columns correspond to the joint outcomes (a, b) :

Table 1: Joint probabilities $p(a, b|x, y)$ for a quantum strategy achieving the Tsirelson bound $\cos^2(\frac{\pi}{8}) \approx 0.8536$.

A	B	$(0, 0)$	$(1, 0)$	$(0, 1)$	$(1, 1)$
a	b	$\frac{1}{2}$	0	0	$\frac{1}{2}$
a'	b	$\frac{3}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{3}{8}$
a	b'	$\frac{3}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{3}{8}$
a'	b'	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

This table illustrates a quantum strategy that violates the CHSH inequality. The marginals are uniform: for each context (x, y) , each outcome (a, b) has probability summing to 1, as required by compatibility. Multiplying by 8 gives an integer counting table at level $t = 2$, which corresponds to one of the lifted PR generators discussed below.

DEFINITION 2.1. A *measurement scenario* is a triple $\langle X, \mathcal{M}, O \rangle$ where X is a set of measurements, O is a set of outcomes, and $\mathcal{M} \subseteq \mathcal{P}(X)$ is a family of measurement contexts. The *event presheaf* associated with this scenario is the functor $\mathcal{E} : \mathcal{P}(X)^{\text{op}} \rightarrow \text{Set}$ defined by $\mathcal{E}(U) := O^U$ for every subset $U \subseteq X$. If $U \subseteq V \subseteq X$, the restriction map $\mathcal{E}(V) \rightarrow \mathcal{E}(U)$ sends a joint assignment $s : V \rightarrow O$ to its restriction $s|_U : U \rightarrow O$. Thus, an element $s \in \mathcal{E}(U)$ is a local assignment of outcomes to all measurements in U . In particular, for a *context* $C \in \mathcal{M}$, an element $s \in \mathcal{E}(C)$ represents a possible joint

outcome for the measurements in C .

Let R be a semiring. For any finite set Y , we define

$$\mathcal{D}_R(Y) := \{d : Y \rightarrow R \mid \text{supp}(d) \text{ is finite}\}.$$

Here Y is only the set on which the distribution is defined; it is not a level. The support is $\text{supp}(d) = \{y \in Y \mid d(y) \neq 0\}$. In this report, $R = \mathbb{R}_{\geq 0}$ gives ordinary non-negative weights, while $R = \mathbb{N}$ gives integer countings.

The **total weight** of d is $|d| := \sum_{y \in Y} d(y)$. For a fixed level $t \in R$, we write

$$\mathcal{D}_R^t(Y) := \{d \in \mathcal{D}_R(Y) \mid |d| = t\}.$$

We also use the graded object

$$\mathcal{D}_R^\bullet(Y) := \cup_{t \in R} \mathcal{D}_R^t(Y).$$

Thus the superscript is the level: $\mathcal{D}_R^t$ means exact level t , while $\mathcal{D}_R^\bullet$ means that the level is not fixed. The argument Y is the underlying set of outcomes. In the probabilistic case one usually fixes $R = \mathbb{R}_{\geq 0}$ and $t = 1$. In the discrete setting one takes $R = \mathbb{N}$ and keeps $t \in \mathbb{N}$ as the number of counted trials.

2.2.1. Empirical models as $\mathcal{D}_R \circ \mathcal{E}$

The event presheaf $\mathcal{E}$ describes which local outcome assignments are available over each set of measurements. To pass from possible assignments to empirical data, one composes it with a distribution or counting functor. Thus the relevant object is not only $\mathcal{E}$, but the composite presheaf $\mathcal{D}_R \circ \mathcal{E}$, defined by

$$(\mathcal{D}_R \circ \mathcal{E})(U) = \mathcal{D}_R(\mathcal{E}(U)).$$

An element $d_U \in \mathcal{D}_R(\mathcal{E}(U))$ assigns a weight to each local section $s : U \rightarrow O$. In particular, a local counting over a context C is an element $N_C \in \mathcal{D}_\mathbb{N}^\bullet(\mathcal{E}(C))$. This means only that N_C has some integer level. If $N_C \in \mathcal{D}_\mathbb{N}^t(\mathcal{E}(C))$, then the context C has exact level t .

In all the following we suppose that $\mathcal{M}$ is a **measurement cover**, meaning $\cup_{C \in \mathcal{M}} C = X$, and that it is an antichain, meaning $C \subseteq D$ with $C, D \in \mathcal{M}$ implies $C = D$.

2.2.2. Compatibility before the level

The first notion is compatibility. It comes from the sheaf-theoretic idea that two local observations must agree on the part that they have in common. This is a local-to-local condition. It does not start by imposing a global normalization constant.

DEFINITION 2.2 (Compatible family). Let R be a semiring. A family

$$e = (e_C)_{C \in \mathcal{M}}, \quad e_C \in \mathcal{D}_R(\mathcal{E}(C)),$$

is **compatible** if, for all contexts $C, D \in \mathcal{M}$,

$$e_C|_{C \cap D} = e_D|_{C \cap D}.$$

In the probabilistic case, this says that marginal distributions coincide on overlaps. In the counting case, it says that marginal count vectors coincide exactly. Equivalently, for a counting family $N = (N_C)_{C \in \mathcal{M}}$ and for $u \in \mathcal{E}(C \cap D)$, compatibility means

$$\sum_{s \in \mathcal{E}(C), s|_{C \cap D} = u} N_{C(s)} = \sum_{r \in \mathcal{E}(D), r|_{C \cap D} = u} N_{D(r)}.$$

We sometimes denote this whole family of homogeneous linear equations by $\delta N = 0$.

2.2.3. From sheaves to hypergraphs

The sheaf-theoretic formulation and the hypergraph formulation describe the same local-to-global problem from two complementary viewpoints. In the sheaf-theoretic approach of Abramsky and Brandenburger, the event presheaf $\mathcal{E}$ assigns to each set of measurements $U \subseteq X$ the set of local sections $\mathcal{E}(U) = \mathcal{O}^U$. Contextuality is then an obstruction to gluing local data into a global section [AB11].

The hypergraph viewpoint keeps the same operational information but makes the linear constraints explicit. Its vertices are the visible events (C, s) . Its hyperedges encode the tests that a table must satisfy: context normalization and marginal compatibility. This is close in spirit to the combinatorial approach to contextuality scenarios developed by Acín, Fritz, Leverrier, and Sainz [Ací+15]. It is also related to the graph-theoretic exclusivity viewpoint of Cabello, Severini, and Winter [CSW14], although here we keep the measurement-cover and marginal-compatibility structure visible.

In this report, the sheaf language explains what must be glued, while the hypergraph and matrix language explains how the gluing constraints act on integer counts.

2.2.4. How compatibility creates the level

For a local counting N_C , define its local level by

$$t_C := |N_C| = \sum_{s \in \mathcal{E}(C)} N_{C(s)}.$$

At first, the numbers t_C may depend on the context C . The point is that compatibility can force them to become equal.

PROPOSITION 2.3 (Compatibility creates the level). *Suppose that the intersection graph of the contexts is connected, where C is adjacent to D when $C \cap D \neq \emptyset$. If a counting family $N = (N_C)_{C \in \mathcal{M}}$ is compatible, then all local levels $|N_C|$ are equal. Hence there is a unique integer t such that*

$$\forall C \in \mathcal{M}, \quad |N_C| = t.$$

Proof. If $C \cap D \neq \emptyset$ and N is compatible, then the marginals of N_C and N_D on $C \cap D$ are equal. Marginalization preserves total weight, so $|N_C| = |N_D|$. Since the intersection graph is connected, this equality propagates from one context to every other context. Thus all local levels are equal to a common integer t . Uniqueness is immediate, because $t = |N_C|$ for any context C .

□

Thus the level is not just an extra parameter added by hand. On a connected cover, it is the common normalization scale forced by compatible integer data.

DEFINITION 2.4 (Discrete empirical model of level t). *A **discrete empirical model of level t** is a compatible family*

$$N = (N_C)_{C \in \mathcal{M}}$$

such that $N_C \in \mathcal{D}_{\mathbb{N}}^t(\mathcal{E}(C))$ for every context C .

2.2.5. Why the level matters

It may look as if the integer model contains no new information, because one can divide a level- t count by t and obtain a probability table. This would miss the main point. The level t is not only a denominator. It records the amount of observation that supports the table in each context.

For instance, the count $(1, 0)$ and the count $(1000, 0)$ give the same normalized probability $(1, 0)$. They do not have the same empirical status. The second table is supported by many more repetitions. In a standard statistical interpretation, this usually means that the sampling uncertainty decreases at the scale $\frac{1}{\sqrt{t}}$. We do not need to assume such a statistical model in this report, but the intuition is useful: t is an experimental resolution scale. It is not a subjective probability of confidence. It is the amount of counted evidence that remains visible in the formal object.

Thus a discrete empirical model is not only a point in a polytope. It is an integer point at a definite level. The same probability may appear at several levels, but these levels correspond to different amounts of evidence and to different possible intermediate histories.

Remark (An arithmetic effect of the level). Consider a partial CHSH count at level 6. Suppose that one context already contains the counts $(3 \ 1 \ 1 \ 1)$. After normalization, this gives probabilities $(\frac{1}{2} \ \frac{1}{6} \ \frac{1}{6} \ \frac{1}{6})$. Now suppose that a marginal compatibility equation involves two unknown probabilities a and b and has the form $\frac{1}{2} + \frac{1}{6} + a + b = 1$. Over probabilities, this gives the continuum of solutions $a + b = \frac{1}{3}$ with $a, b \geq 0$.

At level 6, however, the unknowns must come from integer counts: $a = \frac{A}{6}$ and $b = \frac{B}{6}$ with $A, B \in \mathbb{N}$. The same equation becomes $3 + 1 + A + B = 6$, hence $A + B = 2$. The continuum has been cut down to the finite arithmetic set $(A, B) \in \{(0, 2), (1, 1), (2, 0)\}$.

The point is not uniqueness. The point is that the level cuts the probabilistic continuum by an arithmetic lattice. This arithmetic rigidity disappears after normalization.

This is the reason why the transition to automata is natural. If the level records how much of a process has stabilized, then it is also natural to ask what exists before stabilization. A count of size 3 in CHSH is not a completed level- t model. But it is not meaningless. It may be the visible past of a generator that has not closed yet.

The conceptual chain is therefore the following. The level t describes closed stabilized observations. The Hilbert basis gives elementary closed bricks. Generator automata

describe how these bricks can be produced step by step. Residuals record the future that is still needed for closure. The dynamic model below is designed to keep exactly this missing information: what has already been seen, and what still has to happen for the process to close.

2.2.6. Vector and hypergraph packaging

The vector notation is only a packaging of the sheaf-theoretic definition above. Define the visible event space

$$V := \coprod_{C \in \mathcal{M}} \mathcal{E}(C) = \{(C, s) \mid C \in \mathcal{M} \text{ and } s \in \mathcal{E}(C)\}.$$

Giving a family $(N_C)_{C \in \mathcal{M}}$ is the same thing as giving a vector $N \in \mathbb{N}^V$, by the rule

$$N(C, s) := N_{C(s)}.$$

The context hypergraph is $H_{\mathcal{M}} = (V, E_{\mathcal{M}})$, where

$$E_{\mathcal{M}} = \{e_C \mid C \in \mathcal{M}\} \quad \text{and} \quad e_C := \{(C, s) \mid s \in \mathcal{E}(C)\}.$$

Let $A_{\mathcal{M}}$ be its incidence matrix. Then

$$(A_{\mathcal{M}}N)_C = \sum_{s \in \mathcal{E}(C)} N(C, s) = |N_C|.$$

Therefore the equation $A_{\mathcal{M}}N = t \mathbf{1}$ expresses the common-level condition. It does not express compatibility by itself.

The compatibility equations are homogeneous. If D denotes their matrix, the vector form of a discrete empirical model of level t is

$$A_{\mathcal{M}}N = t \mathbf{1} \quad \text{and} \quad DN = 0.$$

Equivalently, using the notation $\delta N = 0$ for the second family of equations, the compatible counting semigroup is

$$\mathcal{S} := \{(N, t) \in \mathbb{N}^V \times \mathbb{N} \mid A_{\mathcal{M}}N = t \mathbf{1} \text{ and } \delta N = 0\}.$$

For a fixed level t , we write

$$\mathcal{S}_t := \{N \in \mathbb{N}^V \mid (N, t) \in \mathcal{S}\}.$$

If the two families of constraints are assembled into one augmented matrix, then

$$A_{\text{aug}}N = (t \mathbf{1}, 0).$$

The zero part is important: compatibility equations are homogeneous, while normalization equations carry the level t .

2.2.7. From convex mixtures to Hilbert bases

This discrete framework differs from the standard probabilistic one. In the usual convex setting, if e and e' are empirical models, then every convex combination $re + (1 - r)e'$ with $r \in [0, 1]$ is again an empirical model. The geometry is therefore convex.

In our setting, this operation is not primitive. An arbitrary real coefficient r does not preserve integer counts. Instead, the natural operation is addition: if (N, t) and (N', t') belong to $\mathcal{S}$, then

$$(N, t) + (N', t') = (N + N', t + t')$$

also belongs to $\mathcal{S}$. Hence the appropriate algebraic object is not first a convex set, but an affine semigroup.

The analogue of a convex decomposition is therefore a Hilbert-basis decomposition. The Hilbert basis $\mathcal{H}(\mathcal{S})$ of $\mathcal{S}$ is the finite set of irreducible elements of the semigroup. Its elements are the elementary compatible counting models that cannot be decomposed into smaller compatible counting models. Thus, instead of asking whether a model is a convex mixture of deterministic models, we ask which Hilbert generators are required to build it as an integer sum.

2.2.8. Discrete noncontextuality and strong contextuality

A global section is an assignment of an outcome to every measurement at once. Formally, it is an element $g \in \mathcal{E}(X)$, or equivalently a function $g : X \rightarrow O$. The intuitive meaning is simple: before any context is chosen, every measurement $x \in X$ already has a predetermined outcome value $g(x) \in O$.

Thus, a global section represents a deterministic **hidden-variable** assignment. If the experimenter later chooses a context $C \in \mathcal{M}$, the observed local outcome is not chosen anew; it is just the restriction $g|_C : C \rightarrow O$ of the already fixed global assignment.

In other words, determinism means that all counterfactual outcomes are jointly defined. Even if two measurements cannot be performed together, such as a_1 and a_2 in the Bell scenario, a deterministic model still assigns values to both of them in advance.

Given a global section $g : X \rightarrow O$, we define its deterministic counting vector $d_g \in \mathbb{N}^V$ by

$$d_g(C, s) := \begin{cases} 1 & \text{if } s = g|_C \\ 0 & \text{otherwise.} \end{cases}$$

Here $V = \{(C, s) \mid C \in \mathcal{M}, s \in \mathcal{E}(C)\}$ is the set of contextual events. Thus d_g places one unit of mass on exactly one event in each context: the event selected by restricting the same global assignment g to that context.

Equivalently, d_g is the empirical model produced by the deterministic hidden state g . It satisfies the compatibility constraints automatically, because all its local pieces come from one and the same global assignment. In incidence form, $A_{\mathcal{M}} d_g = \mathbb{1}$ and $\delta d_g = 0$; with the augmented convention, this is $A_{\text{aug}} d_g = (\mathbb{1}, 0)$.

More generally, an arbitrary integer counting of global hidden assignments is an element

$$c \in \mathcal{D}_{\mathbb{N}}^{\bullet}(\mathcal{E}(X)).$$

Here $c(g)$ counts how many times the global assignment $g : X \rightarrow O$ is used. Such a global counting produces a contextual-event counting by the linear map

$$\Phi(c) := \sum_{g \in \mathcal{E}(X)} c(+g) d_g \in \mathbb{N}^V.$$

If $|c| = t$, then $\Phi(c)$ has common level t and is compatible. Thus Φ maps global hidden countings into the empirical semigroup $\mathcal{S}$.

Indeed, fix a context C . Since each deterministic vector d_g puts exactly one unit on the event $(C, g|_C)$ and zero on the other events of the same context, we have

$$\sum_{s \in \mathcal{E}(C)} \Phi(c)(C, s) = \sum_{s \in \mathcal{E}(C)} \sum_{g \in \mathcal{E}(X)} c(g) d_g(C, s) = \sum_{g \in \mathcal{E}(X)} c(g) = |c| = t.$$

So every context has the same level t . Compatibility is also automatic. If $C, D \in \mathcal{M}$ and $u \in \mathcal{E}(C \cap D)$, then

$$\sum_{s|_{C \cap D} = u} \Phi(c)(C, s) = \sum_{g|_{C \cap D} = u} c(g) = \sum_{r|_{C \cap D} = u} \Phi(c)(D, r).$$

Both sides count exactly the same global assignments: those whose restriction to the overlap $C \cap D$ is u . Hence $\delta\Phi(c) = 0$.

A discrete empirical model $N \in \mathbb{N}^V$ is **noncontextual** if it can be written as an integer sum of deterministic global sections. That is, there exist coefficients $\{c_g\}_{g \in \mathcal{E}(X)} \in \mathbb{N}$ such that

$$N = \sum_{g \in \mathcal{E}(X)} c_g d_g.$$

The coefficient c_g counts how many times the deterministic assignment g is used in the construction of N .

This is the discrete analogue of the usual hidden-variable decomposition. In the probabilistic setting one asks whether an empirical model is a convex mixture of deterministic assignments. In the counting setting, one asks whether the observed integer table is a finite sum of deterministic counting vectors.

If such a decomposition exists, then the apparent contextual data can be explained by a classical hidden-variable mechanism: each run secretly chooses a global assignment g , and the context merely reveals the part $g|_C$ corresponding to the measurements actually performed.

If no such decomposition exists, the model is contextual in the discrete sense. It is compatible as a counting model, because it satisfies the common-level constraints, but it cannot be reconstructed as a sum of predetermined global assignments.

In the notation above, this says simply that the noncontextual integer models are the image

$$\mathcal{S}_{\text{nc}} := \Phi(\mathcal{D}_{\mathbb{N}}^*(\mathcal{E}(X))) \subseteq \mathcal{S}.$$

This notation is useful because it separates two questions. The semigroup $\mathcal{S}$ contains all compatible integer empirical models. The subsemigroup $\mathcal{S}_{\text{nc}}$ contains only those

compatible models that can be generated from global deterministic assignments. Thus $\mathcal{S}_{\text{nc}}$ is the integer analogue of the classical or local polytope, while $\mathcal{S}$ is the integer analogue of the compatible, or no-signaling, object.

PROPOSITION 2.5. *The semigroup $\mathcal{S}_{\text{nc}}$ is generated by the deterministic vectors $(d_g, 1)$ with $g \in \mathcal{E}(X)$.*

Proof. By definition, every element of $\mathcal{D}_{\mathbb{N}}^{\bullet}(\mathcal{E}(X))$ is a finite counting $c : \mathcal{E}(X) \rightarrow \mathbb{N}$. Hence

$$\Phi(c) = \sum_{g \in \mathcal{E}(X)} c(g) d_g.$$

Therefore every element of $\mathcal{S}_{\text{nc}}$ is an integer sum of deterministic vectors. Conversely, every integer sum of deterministic vectors is obtained by taking $c(g)$ equal to the multiplicity of d_g in the sum. Thus these vectors generate $\mathcal{S}_{\text{nc}}$. □

In the generic case, the Hilbert basis of $\mathcal{S}_{\text{nc}}$ is obtained from these deterministic generators after removing duplicates, because two distinct global assignments may induce the same contextual vector if the measurement cover does not separate them. In the usual separated scenarios, and in particular in CHSH, the deterministic vectors are distinct and irreducible; then the Hilbert basis of $\mathcal{S}_{\text{nc}}$ is exactly $\mathcal{H}_{\text{nc}} = \{(d_g, 1) \mid g \in \mathcal{E}(X)\}$.

Therefore contextuality is a property of a compatible counting $N \in \mathcal{S}$: it means $N \notin \mathcal{S}_{\text{nc}}$. The condition $N \in \mathcal{S}$ alone is not contextuality; it is only compatibility, or no-signaling in Bell scenarios.

We shall only use the support version briefly. For each context C , define

$$\text{supp}(N_C) := \{s \in \mathcal{E}(C) \mid N_C(s) > 0\}.$$

A compatible model N is **strongly contextual** when no global section remains compatible with these supports:

$$\nexists g \in \mathcal{E}(X) \quad \text{such that} \quad \forall C \in \mathcal{M}, \\ g|_C \in \text{supp}(N_C).$$

In this report, this notion is mainly used as orientation. The main constructions below concern compatible countings, noncontextual decompositions, and dynamic generator explanations.

We therefore keep three levels distinct:

- **compatibility** or **non-signaling**: the local count tables have a common level and agree on overlaps, equivalently $A_{\mathcal{M}}N = t \mathbb{1}$ and $\delta N = 0$;
- **noncontextuality**: the compatible model decomposes as an integer sum of deterministic global sections;

- **strong contextuality**: not even one deterministic global section is compatible with the support of the model.

2.3. Small results on the discrete CHSH semigroup

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We now record the elementary consequences of the previous definitions for the CHSH scenario. In this subsection, N denotes the counting vector. The entries of N are the integer variables that we study.

For a context $C \in \mathcal{M}$ and a local section $s \in \mathcal{E}(C)$, the coordinate $N(C, s)$ may also be written $N(s \mid C)$. In the Bell notation this becomes $N(ab \mid xy)$: conditionally on the chosen questions (x, y) , one counts the observed answers (a, b) .

In CHSH the vertex set is $V = \{(ab \mid xy) \mid a, b, x, y \in \{0, 1\}\}$, hence $|V| = 16$. The contextual hypergraph has 12 hyperedges: four context-normalization edges and eight non-signaling marginal edges. Each edge has size 4, and each vertex belongs to exactly 3 edges. If A is the corresponding incidence matrix, then the discrete non-signaling equation is

$$AN = t \mathbb{1}.$$

This is the same object as the non-signaling polytope, but seen before normalization. Indeed, if

$$G_{\text{NS}}(A) := \{p \in \mathbb{R}_{\geq 0}^V \mid Ap = \mathbb{1}\} = P(A),$$

then the equation $AN = t \mathbb{1}$ is equivalent, for $t > 0$, to $p := \frac{N}{t} \in G_{\text{NS}}(A)$. Conversely, any $p \in G_{\text{NS}}(A)$ such that $tp \in \mathbb{Z}^V$ gives an integer model $N := tp$. Thus

$$\{N \in \mathbb{N}^V \mid AN = t \mathbb{1}\} = tG_{\text{NS}}(A) \cap \mathbb{Z}^V.$$

The semigroup $\mathcal{S}_{\text{ns}}(A)$ is therefore the semigroup of integer points in the cone over $G_{\text{NS}}(A)$:

$$\mathcal{S}_{\text{ns}}(A) := \{(N, t) \in \mathbb{N}^V \times \mathbb{N} \mid AN = t \mathbb{1}\} = \{(N, t) \mid N \in tG_{\text{NS}}(A) \cap \mathbb{Z}^V\}.$$

The polytope $G_{\text{NS}}(A)$ is the normalized slice $t = 1$ of this cone, while the discrete models at level t are the integer points in the dilated slice $tG_{\text{NS}}(A)$.

The local part is obtained by replacing $G_{\text{NS}}(A)$ by the local polytope

$$L_{\text{loc}}(A) := \text{conv} \{d_\lambda \mid \lambda \text{ deterministic}\}.$$

Thus

$$Q_{\text{NS}}(t) := \#(tG_{\text{NS}}(A) \cap \mathbb{Z}^V) \quad \text{and} \quad Q_{\text{loc}}(t) := \#(tL_{\text{loc}}(A) \cap \mathbb{Z}^V)$$

count, respectively, all compatible integer non-signaling models and the integer models lying in the local polytope at the same level. The semigroup viewpoint and the Ehrhart viewpoint are therefore two descriptions of the same graded object: $\mathcal{S}_{\text{ns}}(A)$ studies how integer points decompose into irreducible generators, while $Q_{\text{NS}}(t)$ counts how many such points occur in each slice.

PROPOSITION 2.6. *In the CHSH scenario, every discrete model (N, t) satisfies $k = \sum_{v \in V} N_v = 4t$. In particular, there are no discrete models with $k = 1, 2, 3$, nor with any k not divisible by 4.*

Proof. Summing all coordinates of $AN = t \mathbf{1}$ gives $12t$ because CHSH has 12 hyperedges. The same sum also equals $3 \sum_{v \in V} N_v$, because each vertex belongs to exactly 3 hyperedges. Hence $12t = 3k$, so $k = 4t$. $\square$

The first admissible nonzero level is therefore $t = 1$, equivalently $k = 4$. At this level, $AN = \mathbf{1}$ forces N to select exactly one event on every hyperedge. These exact transversals are precisely the local deterministic strategies, i.e. assignments $a = f(x)$ and $b = g(y)$. Thus the first level contains only classical generators.

PROPOSITION 2.7. *The first intrinsically non-local CHSH generators occur at $t = 2$, equivalently $k = 8$.*

Proof. A PR box has probabilities in $\{0, \frac{1}{2}\}$ and therefore is not an integer point at level $t = 1$. Multiplying it by 2 gives an integer vector $r \in \{0, 1\}^{16}$ with $Ar = 2 \mathbf{1}$. Hence its first discrete lift is at level $t = 2$, so $k = 4t = 8$. $\square$

The compatible, or non-signaling, affine semigroup associated with CHSH is

$$\mathcal{S}_{\text{ns}}(\text{CHSH}) := \{(N, t) \in \mathbb{N}^{16} \times \mathbb{N} \mid AN = t \mathbf{1}\}.$$

Let $d_\lambda \in \{0, 1\}^{16}$ denote the 16 local deterministic generators, with $Ad_\lambda = \mathbf{1}$, and let $r_\mu = 2p_\mu^{\text{PR}} \in \{0, 1\}^{16}$ denote the 8 lifted PR generators, with $Ar_\mu = 2 \mathbf{1}$. The natural candidate Hilbert basis is

$$\mathcal{H}_{\text{CHSH}} := \{(d_\lambda, 1)\}_{\lambda=1}^{16} \cup \{(r_\mu, 2)\}_{\mu=1}^8.$$

The noncontextual subsemigroup is

$$\mathcal{S}_{\text{nc}}(\text{CHSH}) := \left\{ \sum_{\lambda=1}^{16} c_\lambda (d_\lambda, 1) \mid c_\lambda \in \mathbb{N} \right\}.$$

Its Hilbert basis is the set of the 16 deterministic generators. The larger non-signaling semigroup $\mathcal{S}_{\text{ns}}(\text{CHSH})$ also contains irreducible non-local generators, namely the lifted PR boxes.

PROPOSITION 2.8. *Assuming the support lemma verified by exhaustive enumeration of the CHSH supports, $\mathcal{H}_{\text{CHSH}}$ is the Hilbert basis of $\mathcal{S}_{\text{ns}}(\text{CHSH})$.*

Proof. The deterministic elements are irreducible because their level is 1. A lifted PR element is also irreducible: a nontrivial decomposition of $(r_\mu, 2)$ would split it into two level-1 elements, hence into two deterministic local models, which would write a PR box as a convex combination of local deterministic boxes. This is impossible. For generation, the support lemma states that every nonzero $(N, t) \in \mathcal{S}_{\text{ns}}(\text{CHSH})$ contains the support of some element $(u, s) \in \mathcal{H}_{\text{CHSH}}$. Then $(N - u, t - s)$ is again in $\mathcal{S}_{\text{ns}}(\text{CHSH})$, and induction on t gives a decomposition into elements of $\mathcal{H}_{\text{CHSH}}$. $\square$

This means that, at the discrete level, CHSH non-signaling models are built from two types of irreducible bricks: local deterministic bricks of level 1 and PR bricks of level 2. The statement depends on the enumerated support lemma; the rest of the proof is formal.

Finally, fixed-level counting is an Ehrhart problem. Let $G_{\text{NS}}(\text{CHSH}) = \{p \in \mathbb{R}_{\geq 0}^{16} \mid Ap = \mathbf{1}\}$ be the CHSH non-signaling polytope and let $L_{\text{loc}}(\text{CHSH})$ be the local polytope. Define $Q_{\text{NS}}(t) := \#(tG_{\text{NS}}(\text{CHSH}) \cap \mathbb{Z}^{16})$ and $Q_{\text{loc}}(t) := \#(tL_{\text{loc}}(\text{CHSH}) \cap \mathbb{Z}^{16})$. Since the affine dimension is 8, these functions have degree 8; Q_{loc} is an Ehrhart polynomial, while Q_{NS} is a quasi-polynomial of period 2 because PR vertices have denominator 2.

A direct interpolation from exact integer-point counts gives

$$Q_{\text{loc}}(t) = 1 + \frac{71}{21}t + \frac{6053}{1260}t^2 + \frac{697}{180}t^3 + \frac{91}{45}t^4 + \frac{13}{18}t^5 + \frac{31}{180}t^6 + \frac{31}{1260}t^7 + \frac{1}{630}t^8$$

and

$$Q_{\text{NS}}(t) = \frac{17}{10080}t^8 + \frac{17}{630}t^7 + \frac{7}{36}t^6 + \frac{37}{45}t^5 + \frac{1607}{720}t^4 + \frac{359}{90}t^3 + \frac{290}{63}t^2 + \frac{332}{105}t + \frac{63}{64} + \frac{(-1)^t}{64}.$$

Thus the number of discrete CHSH models with parameter k is 0 if 4 does not divide k , and is $Q_{\text{NS}}(\frac{k}{4})$ if $k = 4t$. The local polynomial has normalized volume $8! \cdot \frac{1}{630} = 64$, whereas the non-signaling quasi-polynomial has normalized volume $8! \cdot \frac{17}{10080} = 68$. Consequently, the asymptotic non-local part $Q_{\text{NS}}(t) - Q_{\text{loc}}(t)$ has leading term $\frac{1}{10080}t^8$, and the asymptotic proportion of non-local integer models inside the non-signaling ones is $\frac{1}{17}$.

Main concept : dynamic contextual automata

CHAPTER 3

The previous section described compatible countings as elements of an affine semigroup. This is a static description. It tells us which completed integer tables are possible at each level t . The level is important because it records the scale at which the observation has stabilized.

In CHSH, a completed compatible count has total size $k = 4t$. Hence completed objects appear only at sizes divisible by 4. The dynamic question is therefore unavoidable: what can be said after only one, two, or three visible events have occurred? Such a prefix is not a completed empirical model, but it may still be the beginning of a process that will later close into one.

The aim of dynamic contextual automata is to describe these intermediate states without reducing them to noise. A completed count is built from elementary generators. An interrupted count is built from completed generators and from generators that are still open. The open part contains information about the future: it records what must still be emitted in order to close the process. This makes the later definition of a dynamic state almost forced: it must contain closed copies, open copies, and the residual future of each open copy.

The following sections present the result of our work, from the initial definitions and motivations to the final proposition. The first part introduces the foundations of our meta-theory of theories. The second part shows how this framework can account for simple examples of contextuality. We also emphasize the advantage of replacing probabilistic models with an integer-based dynamic model.

3.1. Basic of dynamic contextual automata ---

3.1.1. Generators as complete processes

Let V be the visible event space introduced above. A count is an element of $\mathbb{N}^V$. A generator is a finite counting vector $g \in \mathbb{N}^V$ that is regarded as one elementary complete process.

Generators are not introduced as arbitrary formal objects. They are meant to be elementary bricks of stabilization. A closed empirical table is not explained by a probability point alone, but by a finite production of such bricks.

In CHSH, the previous semigroup computation gives the intended examples. The local deterministic generators are the minimal classical bricks. The lifted PR generators are the minimal non-local bricks of the compatible semigroup. A local deterministic generator has total size 4. A lifted PR generator has total size 8. The automata below do not change these bricks. They only describe their progressive production.

At this stage, we do not need to assume that these are the only possible generators in every scenario. The set of generators is a modelling choice, but it is a constrained choice: it must be rich enough to reconstruct the completed countings that we want to explain, and structured enough to make intermediate states meaningful.

DEFINITION 3.1 (Generator family). *A **generator family** is a finite set*

$$\mathcal{G} \subseteq \mathbb{N}^V$$

*whose elements are called **complete generators**. Each $g \in \mathcal{G}$ has a level $\ell(g) \in \mathbb{N}$ such that $(g, \ell(g)) \in \mathcal{S}$.*

*The family is **closed-complete** for $\mathcal{S}$ if every completed compatible model can be written as a finite sum of generators:*

$$\forall (N, t) \in \mathcal{S}, \quad \exists c : \mathcal{G} \rightarrow \mathbb{N}, \quad N = \sum_{g \in \mathcal{G}} c(g)g \quad \text{and} \quad t = \sum_{g \in \mathcal{G}} c(g)\ell(g).$$

This condition is the closed, semigroup-level requirement. It says that the chosen generators can explain every completed compatible counting. When $\mathcal{G}$ is the Hilbert basis of $\mathcal{S}$, this condition is automatic. In practice, the choice of $\mathcal{G}$ must be made carefully: if it is too small, some counts cannot be explained; if it is too large, the explanation loses structure.

This point is crucial for understanding the framework: the model, or theory, chosen to account for a phenomenon can be more or less fine-grained. The choice of generators is therefore part of the modelling process. A more precise theory requires families of generators with higher levels $\ell(g)$, whereas a less precise theory may only require generators of lower levels.

3.1.2. The automaton of one generator

A generator is complete, but it can be produced in several orders. To model this, we attach a small automaton to each $g \in \mathcal{G}$.

For $v \in V$, let $\varepsilon_v \in \mathbb{N}^V$ be the unit vector at v . Thus $\varepsilon_v(v) = 1$ and $\varepsilon_v(w) = 0$ for $w \neq v$.

DEFINITION 3.2 (Residual automaton of a generator). *Let $g \in \mathcal{G}$. The residual automaton $\mathcal{A}_g$ has states*

$$\{R \in \mathbb{N}^V \mid 0 \leq R \leq g\}.$$

The initial state is $R = g$, and the final state is $R = 0$.

If $R(v) > 0$, there is a transition

$$R \rightarrow R - \varepsilon_v$$

labelled by v . This transition emits the visible event v and decreases the residual part by one unit at v .

The state R is the part of g that remains to be emitted. The emitted part is $g - R$. Thus an interruption of the automaton at residual state R gives a past/future decomposition

$$g = (g - R) + R.$$

The first term is already visible. The second term is the future required to close this copy of the generator.

Remark (A four-event generator). Suppose that $g = \varepsilon_{v_1} + \varepsilon_{v_2} + \varepsilon_{v_3} + \varepsilon_{v_4}$. The automaton $\mathcal{A}_g$ starts at residual state g . If the first three emitted events are v_1, v_2, v_3 , then the residual state is $R = \varepsilon_{v_4}$ and the visible count is $g - R = \varepsilon_{v_1} + \varepsilon_{v_2} + \varepsilon_{v_3}$.

This prefix has size 3. It is not a closed generator. It is nevertheless meaningful, because it knows that the missing future is ε_{v_4} .

3.1.3. Global states and open generators

An experiment can contain several copies of generators at the same time. Some copies are already closed. Other copies are open and have only been partially emitted. The correct hidden state must remember this distinction.

DEFINITION 3.3 (Dynamic state). A *dynamic state* is a pair

$$\eta = (c_\eta, O_\eta).$$

Here $c_\eta : \mathcal{G} \rightarrow \mathbb{N}$ counts the closed copies of each generator. The object O_η is a finite multiset of open copies. An open copy is a pair (g, R) , where $g \in \mathcal{G}$ is the complete generator and $R \in \mathbb{N}^V$ is its residual part, with $0 \leq R \leq g$.

The visible counting associated with η is

$$N_\eta := \sum_{g \in \mathcal{G}} c_\eta(g)g + \sum_{(g, R) \in O_\eta} (g - R).$$

The first sum is the contribution of closed copies. The second sum is the contribution of open copies: each open copy has already emitted $g - R$ and still owes the residual R .

The future residual of the state is

$$R_\eta := \sum_{(g, R) \in O_\eta} R.$$

If no new generator is opened, then closing all current open copies would produce the completed count

$$K_\eta := N_\eta + R_\eta = \sum_{g \in \mathcal{G}} c_\eta(g)g + \sum_{(g, R) \in O_\eta} g.$$

Thus the natural signature of an interrupted state is the pair

$$\Phi(\eta) := (N_\eta, K_\eta).$$

It records both the stabilized visible count (the past) and the completed count (the futur) that would be obtained if all open generators were closed

Remark. There is no reason for the map $\eta \rightarrow N_\eta$ to be injective. Two different hidden states may have the same visible count but different open generators and different residual futures. This is why the dynamic theory cannot be reduced to the visible counting vector alone.

3.1.4. Dynamic explainability

The central question is now not only whether a count is closed, but whether it can be explained as the visible part of an interrupted process.

DEFINITION 3.4 (Dynamic explanation). Let $N \in \mathbb{N}^V$. A **dynamic explanation** of N over $\mathcal{G}$ is a dynamic state $\eta = (c_\eta, O_\eta)$ such that

$$N = N_\eta = \sum_{g \in \mathcal{G}} c_\eta(g)g + \sum_{(g, R) \in O_\eta} (g - R).$$

The set of all such explanations is denoted

$$\mathcal{H}_\mathcal{G}(N) := \{\eta \mid N_\eta = N\}.$$

This formula is the basic accounting rule of the model. A visible count is explained by a number of closed generators and by a multiset of open generators. Each open generator contributes only the part already emitted. Its residual part is not visible yet, but it is part of the hidden state.

DEFINITION 3.5 (Dynamic completeness). Let $\mathcal{A} \subseteq \mathbb{N}^V$ be a class of visible counts that we want to explain. The generator family $\mathcal{G}$ is **dynamically complete** for $\mathcal{A}$ if

$$\forall N \in \mathcal{A}, \quad \mathcal{H}_\mathcal{G}(N) \neq \emptyset.$$

Dynamic completeness is stronger than closed completeness when $\mathcal{A}$ contains prefixes. Closed completeness only explains final compatible models. Dynamic completeness asks that the intermediate counts also come from some population of closed and open generators.

3.1.5. Petri-net reading and concurrency

The previous definitions can be read as a Petri net. A marking is a multiset of tokens. In our case, a token is either a closed copy of a generator or an open copy with a residual state. The marking is therefore the hidden interface of the experiment.

There are three elementary kinds of transitions.

- **Opening:** choose $g \in \mathcal{G}$ and create a new open copy (g, g) .
- **Emission:** if an open copy is (g, R) and $R(v) > 0$, emit the event v and replace (g, R) by $(g, R - \varepsilon_v)$.
- **Closing:** if an open copy has residual 0, replace it by one closed copy of g .

The emission transitions on two different open copies are independent. They can be performed in either order and lead to the same marking. This is the basic Petri-net notion of concurrency: the model does not force a single total order between independent local emissions. The observer may later apply the Parikh compression and

keep only the count N_η , but the hidden marking still remembers which copies are open and what remains to be produced.

PROPOSITION 3.6 (Closed states recover the semigroup). *Suppose that $\mathcal{G}$ is closed-complete for $\mathcal{S}$. If a dynamic state η has no open copy, then $(N_\eta, t_\eta) \in \mathcal{S}$, where*

$$t_\eta := \sum_{g \in \mathcal{G}} c_\eta(g) \ell(g).$$

Conversely, every $(N, t) \in \mathcal{S}$ is represented by at least one dynamic state with no open copy.

Proof. If O_η is empty, then $N_\eta = \sum_{g \in \mathcal{G}} c_\eta(g)g$. Since each $(g, \ell(g))$ belongs to $\mathcal{S}$ and $\mathcal{S}$ is closed under addition, (N_η, t_η) belongs to $\mathcal{S}$. Conversely, closed completeness gives a decomposition of every $(N, t) \in \mathcal{S}$ as a finite sum of generators. This decomposition defines a state with the corresponding closed multiplicities and with no open copy. $\square$

3.1.6. Set of possible explanations

We want to define if $N \subseteq N'$:

$$\mathcal{H}_{\mathcal{G}}(N \rightarrow N') = \{(\eta, \eta') \mid \eta \in \mathcal{H}_{\mathcal{G}}(N), \eta' \in \mathcal{H}_{\mathcal{G}}(N'), \eta \rightsquigarrow \eta'\}$$

with the condition $\eta := (c, O) \rightsquigarrow \eta' := (c', O')$ is equivalent to $\forall g \in \mathcal{G}, c(g) \leq c'(g)$ and $\forall g, \forall R, R', (g, R) \in O \wedge (g, R') \in O' \Rightarrow R(v) \leq R'(v)$. This means that first the second observation don't erase the past of the first observation and has the same possible view of the future.

It means that given two observation, we have a theory of sheaf but about past and futur, we take the union of the past and the intersection of the futur like the restriction of the sheaf.

This proposition explains the relation between the static and dynamic viewpoints. The static semigroup is recovered when all generators are closed. The dynamic model adds the missing layer: it also describes states that occur before closure.

3.1.7. First interpretation

The model separates three levels.

- A **trace** is an ordered sequence of visible events produced by hidden transitions.
- A **count** is the Parikh compression of a trace. It forgets the order and keeps only multiplicities.
- A **dynamic state** is a hidden explanation of a count by closed and open generators.

This distinction is essential. A completed count such as an element of $\mathcal{S}_{\text{ns}}$ is only the endpoint of a process. An interrupted experiment is described by an interface η , or equivalently by its past/future signature $\Phi(\eta) = (N_\eta, R_\eta)$. The visible past says what has already happened. The residual future says how the process can still stabilize.

The next step will be to use this interface structure to compare interrupted states, to define costs of open non-local resources, and to study which choices of generators give a meaningful reconstruction of contextual phenomena.

*Analyse quantum experiences
result with dynamic contextual
automata*

CHAPTER 4

Towards a dynamic condition to capture quantum non-locality

CHAPTER 5

Appendix

CHAPTER 6

6.1. The initial state of the art and the initial subject —

We start from a now central observation in quantum complexity theory: entangled multi-prover interactive proofs are extremely powerful, as captured by the equality $\text{MIP}^* = \text{RE}$. In the standard MIP^* model, a verifier interacts with spatially separated provers that may share arbitrary quantum correlations. Importantly, the model does not impose any explicit structural constraint on how entanglement is distributed, nor on what information is operationally accessible to subsets of provers. This abstraction is theoretically useful, because it isolates the informational consequences of quantum non-locality. However, it also incorporates a strong implicit assumption: the availability of unstructured, globally coordinated correlations at no explicit cost.

In physically motivated settings, information propagation is constrained by locality. Interactions are limited by distance and propagation time, and causal structure bounds what can be coordinated within a given time window. Even when large-scale entangled resources are assumed to be pre-distributed, the ability to access and coordinate arbitrary distant subsystems is not free, since synchronization and correlation across long distances become resources in their own right.

The first idea of the interphip was to study the models that retain quantum non-locality while enforcing explicit locality constraints, whether geometric, combinatorial, or causal. The goal is not to weaken MIP^* a priori, but to identify which aspects of its power rely on unconstrained global correlations, and which persist under structured, bounded access to entanglement.

To this end, we would have liked to introduce a localized variant of entangled multi-prover interactive proofs, denoted $\text{MIP}_0^*(k, n)$. The definition separates two levels. First, a global resource specifies how entanglement is distributed, for instance via a large entanglement graph state. Second, each effective interaction is constrained to exploit only local correlations, represented by induced subgraphs of size at most k . Informally, at any point in the protocol, only a “local group” of at most k provers may access and exploit the correlations supported on a k -vertex window of the global resource, as if the verifier could probe only bounded-size subsystems. To preserve the flexibility of multi-prover protocols, we allow n independent copies of the resource state, enabling repeated use of the same local structure across many interactions, while preventing the protocol from implicitly assuming a single globally accessible instance with unbounded coordinated correlations.

This model aims to capture a simple principle: quantum non-locality is not removed, but organized. The parameter k acts as a correlation granularity, or an effective locality radius, bounding the entanglement that can be operationally mobilized within one

interaction. The parameter n functions as a parallelism budget, modeling the ability to distribute multiple independent instances of the same resource without obtaining unconstrained global correlations inside a single instance. Consequently, any global power must arise from repetition, composition, and consistency constraints across local windows, rather than from immediate access to arbitrary global entanglement.

The framework supports two complementary objectives. First, it provides a language for threshold questions, such as determining for which growth rates of k (as a function of the input size) the model recovers the full power of MIP^* , and under which restrictions it yields strictly weaker classes. Second, it creates a common formal setting connecting tools from nonlocal games, entanglement graph theory, and distributed models, because the “induced subgraph of size k ” constraint enforces an inherently local viewpoint compatible with notions of radius, distance, and composition. We next discuss how ideas from distributed computing offer a concrete methodology for reasoning about bounded information access.

To do so, i first read four papers [BW24], [Bal+24], [Man+25] and [Cui+25]. The first goal was to find relevent link the can motivate our approch.

6.1.1. Locality and quantum advantage in distributed computing

The paper [Bal+24] exhibits a separation between classical and quantum distributed computation for a problem specified purely by local constraints. The authors define an LCL problem in a bipartite black-white formalism, where white nodes represent vertices and black nodes represent edges, allowing uniform constraints to be expressed via multisets of input-output pairs. They introduce the *iterated GHZ* problem, which can be viewed as a local composition of GHZ-type nonlocal games whose consistency is locally checkable. Their main result shows that on graphs of maximum degree Δ , any classical strategy requires $\Omega(\Delta)$ rounds, whereas a quantum strategy achieves a constant number of rounds by locally distributing GHZ states and then performing measurements without further communication.

Methodologically, the key point for our purposes is how bounded radius induces structured uncertainty. The round elimination technique transforms a problem Π_i into a derived problem Π_{i+1} whose labels are sets of labels of Π_i . This “set lifting” encodes exactly what a node can no longer distinguish when one communication round is removed. Since this transformation quickly becomes intractable at the level of explicit descriptions, the authors introduce a sequence of relaxations that preserves the operational implication: solving a relaxation in t rounds implies solving the preceding problem in $t + 1$ rounds.

Conceptually, this provides a template for our program. Instead of relying on implicit global coordination, one explicitly models the information available within a bounded neighborhood, replaces the inaccessible part by a structured description of possibilities, and studies how local constraints compose to enforce global correctness. This perspective suggests viewing locality as a constraint on distinguishability, and therefore as a constraint on the representation of information. One can think of an abstract “information band” whose position corresponds to the level of accessible structure. Moving toward smaller locality windows corresponds to replacing precise facts by sets of feasible possibilities consistent with local observations, which increases descriptive un-

certainty and forces an expansion of the local alphabet. Moving toward larger windows corresponds to reducing uncertainty by enforcing compatibility constraints, effectively projecting ambiguous descriptions onto coherent ones. In distributed settings, this is exactly what the transformation $\Pi \mapsto R(\Pi)$ formalizes: reducing the communication radius increases uncertainty, which must be compensated by richer local descriptions.

In our localized MIP* framework, an analogous mechanism arises from bounding the exploitable entanglement. We assume a global resource, such as an entanglement graph, but restrict each interaction to induced subgraphs of size k , possibly across n independent copies. Decreasing k forces most of the global structure out of view, which suggests that any soundness or completeness analysis must work with coarser, set-valued descriptions of what distant provers could be doing, whereas increasing k allows more direct constraints and reduces ambiguity. Our goal is to formalize this trade-off, identify which combinatorial properties of the resource graph control the power of $\text{MIP}_0^*(k, n)$, and determine how locally accessible quantum correlations can compose to enforce global language recognition.

6.1.2. Combinatorial characterizations of quantum games

A recurrent thing is that operational properties of quantum protocols can be equivalently expressed as structural properties of purely combinatorial objects. Several results in the literature illustrate this principle with remarkable precision.

In [BW24], the authors study the isomorphism game: given two graphs $\mathcal{G}$ and $\mathcal{H}$, if the graphs are isomorphic, then the game admits a perfect classical strategy. By extension, they define the notion of quantum isomorphism $\mathcal{G} \cong_q \mathcal{H}$ and non-signaling isomorphism $\mathcal{G} \cong_{\text{ns}} \mathcal{H}$, which hold whenever there exists a perfect quantum or non-signaling strategy, respectively. A key structural result is that non-signaling isomorphism is equivalent to fractional isomorphism, that is, $\mathcal{G} \cong_{\text{ns}} \mathcal{H}$ if and only if there exists a bistochastic matrix u such that $A_{\mathcal{G}}u = uA_{\mathcal{H}}$. This is a clean relaxation of the classical notion of isomorphism, which requires u to be a permutation matrix.

A first striking result, from [MR20], establishes that $\mathcal{G} \cong_{\text{ns}} \mathcal{H}$ is equivalent to $\forall \text{ tree } \mathcal{K}, \# \text{Hom}(\mathcal{K}, \mathcal{G}) = \# \text{Hom}(\mathcal{K}, \mathcal{H})$. This is a beautiful correspondence because the two sides are very different in nature: on one side stands a game, which is a protocol with interactions, rules, and strategies, and on the other side stands a simple combinatorial property involving homomorphism counts. Even more surprisingly, an analogous equivalence holds for quantum isomorphism: $\mathcal{G} \cong_q \mathcal{H}$ if and only if $\forall \text{ planar } \mathcal{K}, \# \text{Hom}(\mathcal{K}, \mathcal{G}) = \# \text{Hom}(\mathcal{K}, \mathcal{H})$. The fact that the laws of quantum mechanics can be captured by a condition on planar homomorphism counts is a striking illustration of the power of combinatorial characterizations. This result is precisely the kind of equivalence we want to find.

A second important result from the same line of work is the construction of the notion of D -fractional isomorphism. The authors introduce a D -distance game that imposes that the output node of both players must be at the same distance from the input node, but only when the distance is below D . This forces a kind of locally consistent isomorphism. The key observation is that this local condition $\mathcal{G} \cong_{\text{ns}}^D \mathcal{H}$ enforces a global

property $\mathcal{G} \cong_{\text{frac}} \mathcal{H}$. This is directly relevant to our programme: it demonstrates that a local condition can enforce a global property.

6.1.3. Quantum perfect matchings

The paper [Cui+25] studies a family of nonlocal games built around the notion of perfect matching. In the bipartite L -perfect matching game BPM_G , the verifier asks questions corresponding to vertices on one side of a bipartite graph, and the players answer with edges. The winning conditions are local: each answer must be incident to the queried vertex, and two answers must be compatible with being part of the same matching. Classically, having a perfect strategy is equivalent to the existence of an actual perfect matching.

What interested us in this paper is precisely this passage from local tests to a global combinatorial object. The verifier never sees the whole matching directly; it only checks local consistency conditions. Nevertheless, in the classical case these local checks force the existence of a global perfect matching. This is close to the kind of mechanism we were looking for when trying to build a toy model for my tutor's broader intuition: global structure should emerge from local compatibility constraints.

The quantum case gives a richer picture. For some versions of the game, quantum strategies do not give an advantage over classical ones: in particular, the bipartite L -perfect matching game is quantum sound. But in more general settings, the paper introduces a genuine notion of quantum perfect matching. The main characterization states that a graph G has a quantum perfect matching if and only if its line graph $L(G)$ admits a projective packing of the appropriate value. In this formulation, edges of the original graph are replaced by projectors, and the condition that two incident edges cannot both be selected becomes an orthogonality condition.

This result was useful for us as a methodological example. It shows that a nonlocal game can sometimes be translated into a clean algebraic-combinatorial condition on a derived graph. The line graph is not just a technical construction: it records which local choices are incompatible. This suggested a general way of thinking about local quantum protocols: instead of trying to understand the whole quantum strategy directly, one can look for a graph encoding the conflicts between local constraints, and then study projectors, orthogonality, and packing-type conditions on this graph.

The paper also contains an important warning. When the same ideas are extended to hypergraphs, deciding the existence of quantum perfect matchings becomes undecidable. This shows that local quantum constraints can quickly become extremely expressive.

6.1.4. Gap-preserving reductions for entangled-prover games

The paper [Man+25] studies gap-preserving reductions between entangled-prover games. Its main question is how to transform one nonlocal game into another while preserving the distinction between perfect strategies and strategies that are bounded away from perfect. This is important because, without control of the gap, a reduction may preserve exact satisfiability but lose all useful quantitative information.

The authors show that general synchronous games can be reduced to independent set games with only polynomial loss in the gap. Given a synchronous game $\mathcal{G}$, they build a game graph $X(\mathcal{G})$ whose vertices are question-answer pairs (q, a) . Two vertices are adjacent when the corresponding answers would make the players lose. A perfect strategy for the original game then corresponds to a suitable independent-set-type strategy on this graph. The main theorem shows that if

$$\omega^*(\mathcal{G}) = 1,$$

then the reduced independent set game also has value 1, while if

$$\omega^*(\mathcal{G}) \leq 1 - \varepsilon,$$

then the reduced game has value at most

$$1 - \Omega\left(\frac{\varepsilon^8}{t^4}\right),$$

where t is the number of questions. Thus the reduction preserves the gap up to a polynomial loss.

The technically difficult part is the approximate case. A near-perfect quantum strategy only gives relations that are approximately true: measurements are almost projective, orthogonality is almost satisfied, and consistency is only approximate. The paper proves a stability theorem showing that such approximate measurement families can be rounded to genuine projective measurements without losing too much. This is the core mechanism that makes the reduction gap-preserving rather than merely exact.

This paper was relevant to us because it gives a precise example of how local combinatorial constraints can retain the hardness of a much more general quantum game. The independent set game has a very simple rule: equal questions require equal answers, and different questions require non-adjacent answers. Yet, through the game graph construction and the stability theorem, this simple game family can encode the complexity of general entangled-prover games.

6.1.5. Conclusion

For our internship, the initial motivation was not only complexity-theoretic. My tutor had developed a broader theoretical viewpoint which seemed to become more stable and coherent over time. After many discussions, I became interested in building a toy model that could justify or at least motivate this way of seeing information, structure, and quantum behaviour. In that context, the paper was useful less as a direct roadmap toward a new complexity class, and more as a proof that local compatibility rules, when organized correctly, can preserve deep global properties. The key ideas we retained were the construction of a derived graph from a game, the role of approximate-to-exact stability, and the importance of controlling how errors accumulate when local constraints are combined.

6.2. Some genaral definition

We recall the standard definition of multi-prover interactive proof systems with entanglement. The parameters are the number of provers r , the number of rounds p , the completeness parameter c , and the soundness parameter s .

Definition 1 ($\text{MIP}_0^*(r, p)$). A language $\mathcal{L}$ belongs to $\text{MIP}^*(r, p)$ if there exists a polynomial-time verifier V interacting with r non-communicating provers $P_1, \dots, P_r$ over p rounds of communication, such that:

1. **Shared entanglement.** Before the protocol begins, the provers share an arbitrary quantum state

$$|\psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \dots \otimes \mathcal{H}_r$$

where each $\mathcal{H}_i$ is a Hilbert space of *arbitrary finite dimension*. No restriction is placed on $|\psi\rangle$: it may be entangled in any way, in any dimension.

2. **Strategy.** Upon receiving question q_i from the verifier, prover P_i performs a measurement $\{M_{a|q_i}^{(i)}\}_a$ on their local subsystem $\mathcal{H}_i$ and returns the outcome a . The joint probability of outcomes $(a_1, \dots, a_r)$ given questions $(q_1, \dots, q_r)$ is

$$p(a_1, \dots, a_r \mid q_1, \dots, q_r) = \langle \psi \mid M_{a_1|q_1}^{(1)} \otimes \dots \otimes M_{a_r|q_r}^{(r)} \mid \psi \rangle.$$

3. **Completeness.** If $x \in \mathcal{L}$, there exist a state $|\psi\rangle$ and measurements such that $\Pr[V \text{ accepts}] \geq 1$ (we have $c = 1$).
4. **Soundness.** If $x \notin \mathcal{L}$, for *every* state $|\psi\rangle$ (of every dimension) and *every* choice of measurements, $\Pr[V \text{ accepts}] \leq \frac{1}{2}$ (we have $s = \frac{1}{2}$).

The class MIP_0^* is defined as $\bigcup_{r,p} \text{MIP}_0^*(r, p)$, allowing arbitrarily many provers and rounds with perfect completeness and bounded-error soundness.

In the MIP^* class, instead of fixing the completeness and soundness parameters to 1 and $\frac{1}{2}$, we take unions over all c, s such that $c - s \geq \frac{1}{\text{poly}(|x|)}$, allowing for arbitrary completeness-soundness gaps.

6.2.1. Two-prover, one-round characterization

A fundamental simplification result shows that two provers and a single round suffice.

Theorem 1 (see [Ji+22]). $\text{MIP}_0^* = \text{MIP}_0^*(2, 1)$.

That is, every language recognizable by an entangled multi-prover interactive proof system is already recognizable by a *two-prover, one-round* protocol with perfect completeness and soundness $1/2$. In this simplified model, the protocol consists of a single exchange:

1. The verifier samples a pair of questions (q_A, q_B) and sends q_A to prover A , q_B to prover B .
2. The provers, who share $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$, each measure their subsystem and return answers a, b .
3. The verifier evaluates a predicate $V(q_A, q_B, a, b)$ and accepts or rejects.

This is precisely the framework of a *nonlocal game*, and the supremum of the acceptance probability over all quantum strategies is the *entangled value* ω^* of the game.

6.2.2. The role of unbounded dimension in $\text{MIP}^* = \text{RE}$

The following is the landmark result that motivates our investigation.

Theorem 2 (Ji, Natarajan, Vidick, Wright, Yuen [Ji+22]). $\text{MIP}^*(2, 1, 1, s) = \text{RE}$ for some constant $s < 1$.

The proof constructs, for each Turing machine M , a nonlocal game $\mathcal{G}_M$ such that:

- if M halts, then $\omega^*(\mathcal{G}_M) = 1$ (a perfect strategy exists in finite dimension d , where d depends on the halting time of M);
- if M does not halt, then $\omega^*(\mathcal{G}_M) < 1$ (no strategy in any dimension achieves value 1).

A crucial feature of this construction is that the entangled value is defined as

$$\omega^*(\mathcal{G}) = \sup_{d \geq 1} \sup_{|\psi\rangle \in \mathbb{C}^d \otimes \mathbb{C}^d} \sup_{\{M_{a|q}\}, \{N_{b|q'}\}} \Pr[\text{win}],$$

and the supremum over d is *essential*. For a halting machine with halting time T , the dimension d of the optimal strategy may grow as a function of T , and no uniform bound on d exists across all instances. In other words, the full power of $\text{MIP}^* = \text{RE}$ relies on the provers' ability to use shared entanglement of *arbitrarily large dimension*.

Remark. If the dimension is bounded by a fixed constant d , the resulting class $\text{MIP}^*(2, 1, s, d)$ is known *not* to capture all of RE ; indeed, for fixed d , the entangled value $\omega_d^*(\mathcal{G})$ is computable by semidefinite programming, so the class collapses to a decidable complexity class.

Bibliography

CHAPTER 6.2.2

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